NaCl-H_2O System (1:62)

"Physical Meanings of RDF Curves" (To get C.N. in a specific region, say $r \in (r_1, r_2)$,
$$N = 4\pi\rho \int_{r_1}^{r_2} r^2 g(r)\,dr$$)

① **O-H RDF Curve**:

→ 1st peak: A sharp peak, centered @ $r \sim 1Å$: @ 330K (using SCAN), the OH bond length is almost 1Å.
So, there will be a peak expected @ $r \sim 1Å$

→ Dip: ∃ a dip, s.t. rdf = 0 after 1Å. Why? So for the given setup, & water molecules cannot come closer than a certain distance, ∴ the rdf has a value = 0 for the distances slightly more than 1Å.

→ 2nd Peak: The second peak represents the hydrogens which are bonded via H-bonds.

→ 3rd Peak: Slightly taller than the 2nd peak! This happens ∵ the 2nd peak comes from the Hydrogens of the second shell as well as those hydrogens of the first shell (the H-bonding Hydrogens ~~2nd~~ shell) which didn't form H-bond w/ central O.

② **O-O RDF**: The first peak of the O-O RDF comes after the ~~first~~ 2nd peak of OH curve.
This happens ∵ the electronegativity difference b/w O and H make the water orient in a way ~~so~~ that H is closer to O than O.

③ **O-Na RDF**: The first peak of O-Na comes after the 2nd peak of OH & before the 1st peak of O-O.
This happens ∵ O & Na^+ will come close to each other due to electronegativity difference, but, the closeness is not as much as it is for O-H. This happens ∵ the size of Na^+ is more than H and by Pauli's exclusion principle, they cannot come closer than OH.

④ **O-Cl RDF**: By the similar intuition, we can say that O-Cl will be far, and they are. But not very far, and the peak isn't very diffused, why?
This happens because Cl^- ion can also form an H-bond with Hydrogens.
however, the bond is weaker due to the following reasons:

→ the charge is spread on a larger Cl^- ion; making the density lower.

→ Water forming H-bonds which are directional. But for Cl^- ion; $\nexists$ any directionality because for HOH: 

. The lone pairs give it an almost tetrahedral shape; making the H-bond directional.
But to make H-bond w Cl^-; H can approach from anywhere, there's no directionality. (real H-bonds involve CT & when we have e^-'s in one dirⁿ; the CT is much stronger as compared to an isotropic distribution)

e) Na-Cl RDF: The first peak shows the presence of peak @ $r = r_{Na^+} + r_{Cl^-}$, which is the peak corresponding to the direct contact. Now; since direct contact isn't a very frequent thing; the more frequent is the contact b/w hydrated ions; and hence the second peak is taller!